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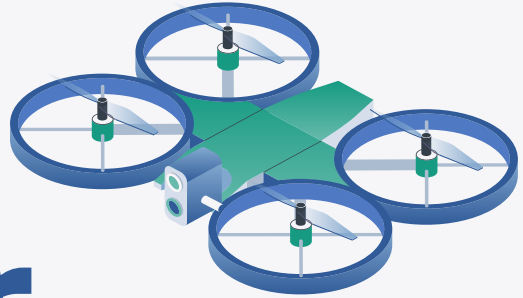

Fall Semester Report

BAE SYSTEMS

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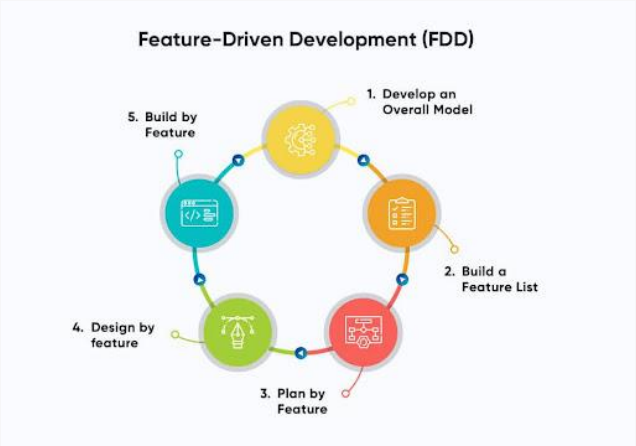
01

Project Methodology

Feature Driven Development

Activities

1. Backlog Refinement - One a month
2. Feature Planning - Once per feature
3. Feature Design Meeting - Twice per feature
4. Standup Meeting - Every Tuesday



Artifacts

1. Overall system model - Done once
2. Feature List - Done once, modified as needed
3. Design - Once per feature
4. Plan - Once per feature
5. Milestone - One each week
6. Incremental Prototypes - Once a month

Metrics

Machine Learning

Precision
Recall
F1-Score
Accuracy
mAP@0.5

Performance

Image poll rate
Image processing rate
AR FPS
Drone flight speed

Process

Story point completion
Full story point completion

Product

Feature slippage
Simultaneous drone count

02

Requirements

AR Requirements

AR Sandtable

- The sandtable shall display local topography
- The sandtable shall show detections on the map
- The sandtable shall notify AR users of new detections
- The sandtable shall be synchronized for all AR users
- The sandtable shall visually display the the age of detection data

AR Synchronization

- The AR glasses shall display the sandtable with the same data at the same time
- The AR glasses shall display the same popups at the same time
- The AR glasses shall display the same detection pins at the same time

Data Processing Requirements

Automated Drone Flight

- The drones shall fly in a predetermined flight pattern
- The drones shall report data such as video and location to the onsite laptop
- The drones shall have a manual override to allow for user control

Automated Dog Detection

- The pop ups shall include location data of the detection
- The pop ups shall include chip-out images of detections
- The pop ups shall allow for user verification
- The model shall take in frames of imagery from the drones
- The model shall detect dogs in imagery frames
- The model shall draw bounding boxes around detected targets
- The model shall report confidence scores for each detection
- The model shall crop imagery so that a new chipout is generated for each detection

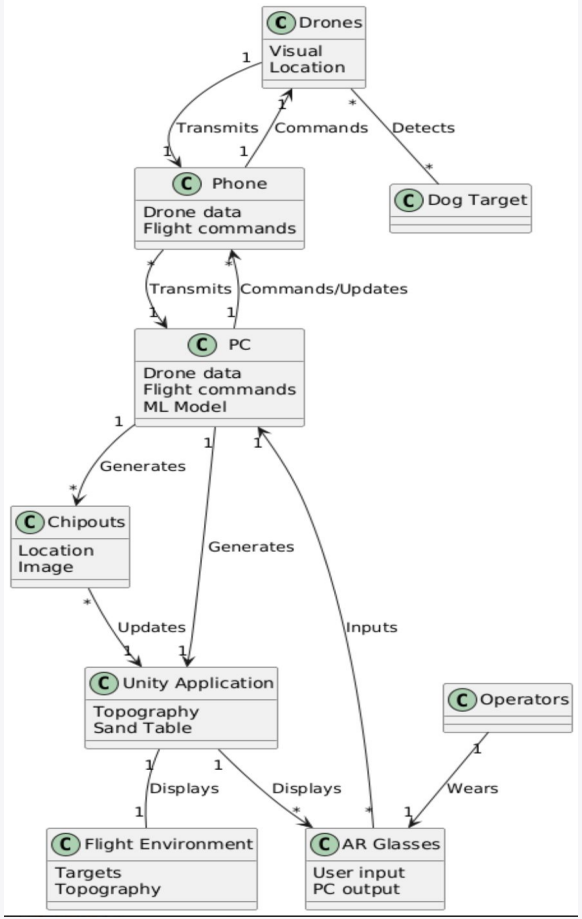
03

Architecture

Architecture

Services

- Drone navigation manager
- Drone image feed processor
- Machine learning processor
- AR rendering service
- Detection update manager



04

Implementation

Point Clouds

Terrain Generation

- Point clouds are being used for generating the current terrain. Our plan is to update/build upon the previous generated points as the drone flies overhead, reducing the generation time

Pre-Trained Models

- Our approach is using pre-trained depth models to estimate the depth within the image. These pre-trained models are trained on 2D images and their 3D model counterparts
- The pre-trained model reviews how light is projected onto the 2D image and estimates a depth, then determines the accuracy based off of the corresponding 3D model counterpart

Challenges

- Determining depth from a single 2D image can be inaccurate especially when dealing with lighter colors
- Larger quantity of updates to previously generated terrain
- Different pre-trained models handle certain environments better than others, such as indoor and outdoor environments or environments requiring deeper depth

Point Cloud Trial

Drone Image (Top Down View)

Point Cloud (Top Down View)

Point Cloud (Side View)

Digital Elevation Model (DEM)

DEM Overview

- DEMs are a 3D model that displays height maps of the terrain in a given area. Many DEMs are freely available on state and local levels which we will use to create the AR sandtable

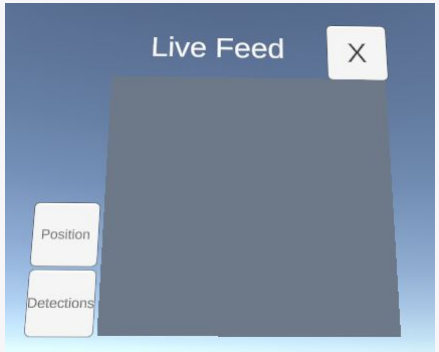
Loading Into AR

- DEMs can be brought into Unity by first converting the elevation data into a heightmap using an external tool such as GDAL. After conversion, the heightmap can be saved as a texture and imported into Unity's Terrain system, where it generates a detailed 3D landscape.

AR Menus

Position menu to move the drone to a specific location

Detection menu that displays the most recent detections with their respective chip-outs



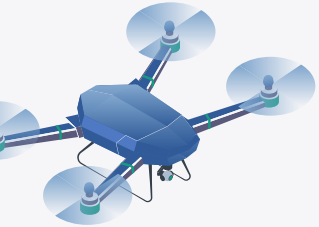
Live feed menu that will display the live video from a specific drone

*Does not include live data but rather the skeleton where pieces of data will be inserted

Test Flight Accomplishments

Tested Android App

The drone to laptop image pipeline was successful tested while flying

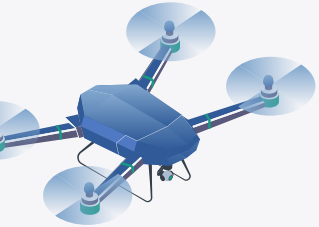


Gathered Image Data

The image data we gathered will be useful for initial **point cloud generation prototypes**

* No bystanders were on or near the playground when the drone was flown over

Test Flight Accomplishments (continued)



Automated Flight Pathing

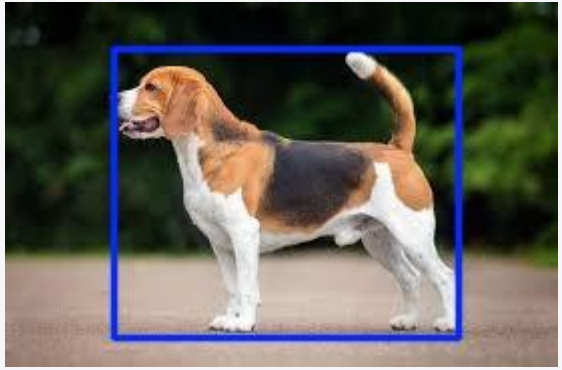
Both the automated flight control and drone to laptop image pipeline were successfully tested.

Initial Model Detections

Initial Ground Model

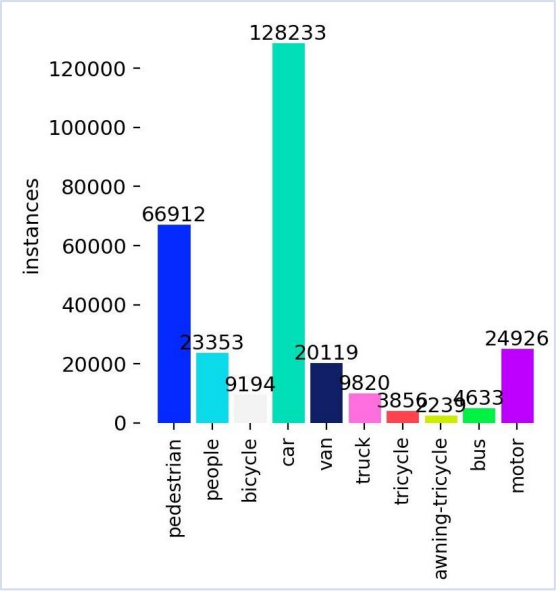
- Our first model was a proof of concept that could detect dogs on the ground
- The model would draw bounding boxes and crop images to show the detected dogs
- While a good demonstration, the model could not handle aerial angles

Full sized ground image with bounded detection



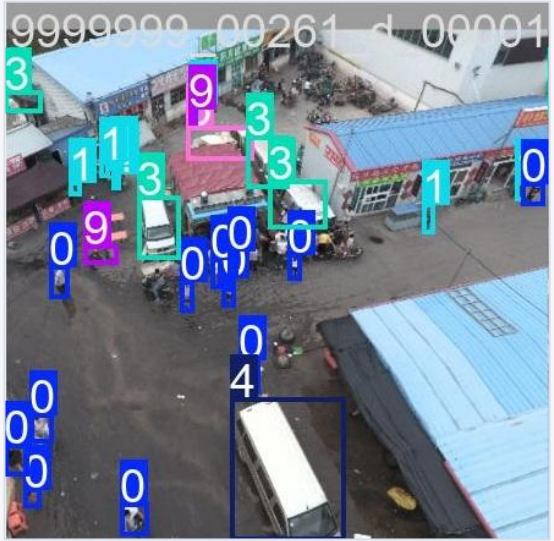
Chipout for AR display showing the detection

YOLOv8 Training on Visdrone



Number and types of detections predicted during one, of many, rounds of training

Predictions done during training on a busy image



Aerial Model Training

- Baseline YOLOv8 model trained on COCO dataset
- To teach the model how to detect from the air, we trained it further on the Visdrone dataset
- This complex dataset gives us a strong foundation to begin injecting our own data focusing on dogs

Current Model In Practice

Current Aerial Model

- After training, we ran our model on data gathered from our test flight
- Results show that the model is doing a good job of doing detections and drawing bounding boxes
- Chipouts are created with the same process as our proof of concept model and will serve as the “thumbnail” for the AR detection display

New Training Data For Aerial Model

Difficult Frames

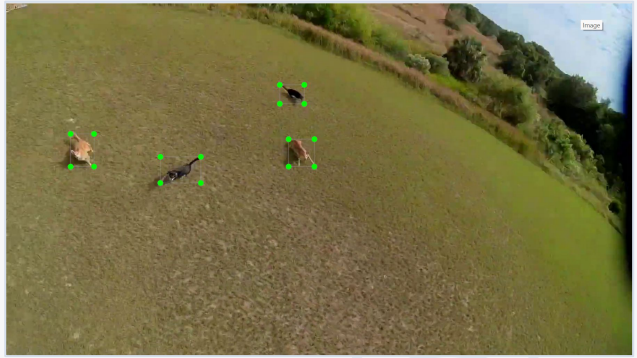
Some training data will include very small images of dogs from a distance

Data Infusion

Our model can see from the air, but it now needs to be taught how to detect dogs specifically

Balanced Data

Creating varied annotated training data of dogs from a top down perspective will yield a stronger final model



Easy Frames

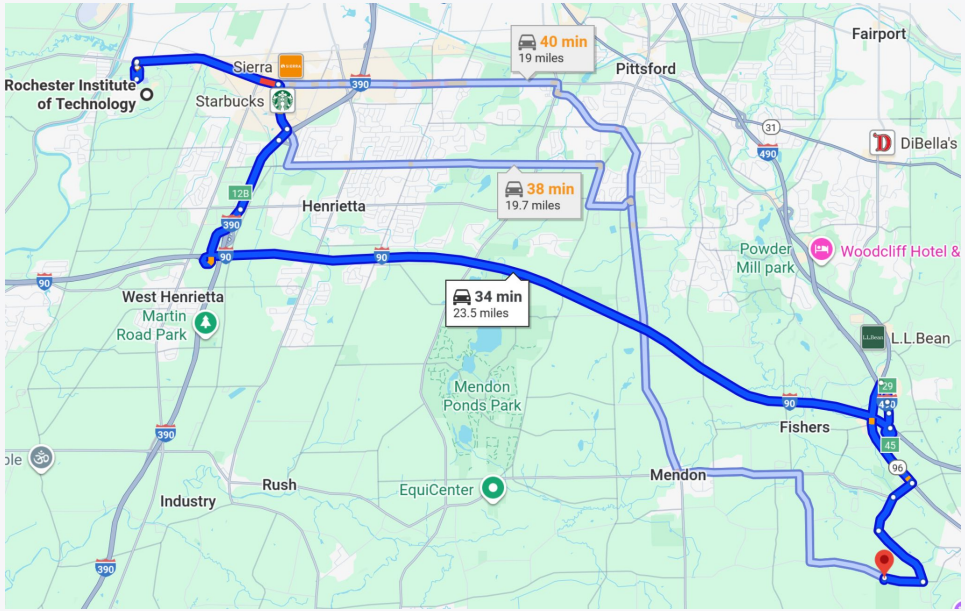
Some training data will include very clear and close images of multiple dogs

05

Testing

Testing Challenges

- Airspace Restrictions
- Monroe County Restrictions
- Continuity of Testing
- Unit Test Constraints

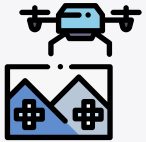


06

Future Plans

Spring Semester Milestones

Key
On-Time
At Risk of Delay
Delayed

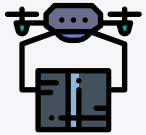


January

- Finalized Data Flow Pipeline
- AR Synchronization
- Multi-drone support for automated flight pathing

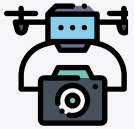
February

- AR Live Video View
- Dynamic Detection Pop Ups



March

- Manual Drone Camera Control
- Individualized AR Controller Menus



April

- Finalize AR Operator Views
- Manual Drone Position Control

Thank You